

7.1.

Let $\{f_n: X \rightarrow Y\}$ be a uniformly convergent sequence of bounded functions.

Prove it is uniformly bounded.

Proof. Given $\epsilon > 0$, there is $N \in \mathbb{N}$ s.t. $n \geq N \Rightarrow \sup_x |f_n - f| \leq \epsilon$, where f is the limit function of f_n .

Therefore, $\sup_x |f| \leq \sup_x |f_N| + \epsilon$, the limit function f is bounded. Now,

for each $1 \leq i < N$, let M_i be a bound of f_i . Then, for all $k \in \mathbb{N}$,

$$\sup |f_k| \leq \max\{M_1, \dots, M_{N-1}, \sup |f| + \epsilon\}.$$

7.2-3. Let $f_n, g_n: X \rightarrow Y$ be functions.

1). If $\{f_n\}, \{g_n\}$ converges uniformly, then $\{f_n + g_n\}$ converges uniformly.

2). If $\{f_n\}, \{g_n\}$ converges uniformly of bounded functions, then $\{f_n g_n\}$ converges uniformly.

3). Find Counter example for 2), the boundedness omitted.

Proof. 1) can be shown directly, by $\sup |f_n + g_n| \leq \sup |f_n| + \sup |g_n|$.

But 2) is needed the boundedness condition for example,

$$f_n: [0, 1] \rightarrow \mathbb{R}: x \mapsto \frac{n}{n+1} \cdot \frac{1}{x} \quad \left(\sup |f_n - \frac{1}{x}| = \frac{1}{n+1} \rightarrow 0 \right)$$

$$g_n: [0, 1] \rightarrow \mathbb{R}: x \mapsto \frac{n+1}{n^2} \cdot \frac{1}{x} \quad \left(\sup |g_n - \frac{1}{x}| = \frac{1}{n} + \frac{1}{n^2} \rightarrow 0 \right)$$

But $f_n g_n(x) = \frac{1}{nx}$ is not uniformly convergent, because $\sup |f_n g_n - 0| = \infty$, but $f_n g_n \rightarrow 0$.

Now suppose that f_n and g_n are bounded (of course, uni. convergent.)

By the above problem, f_n and g_n are uniformly bounded. Now,

$$|f_n g_n - f g| = |(f_n - f) \cdot g_n + (g_n - g) \cdot f| \leq |g_n| |f_n - f| + |f| |g_n - g|$$

gives the result, (More specifically, given $\epsilon > 0$, there is $N \in \mathbb{N}$ s.t.

$$n \geq N \Rightarrow \sup |f_n - f|, \sup |g_n - g| < \epsilon.$$

Therefore, for these $n \geq N$,

$$\sup |f_n g_n - f g| \leq 2M\epsilon, \text{ where } M \text{ is bound of } |g_n| \text{ and } |f|.$$

7.4. Consider a map defined on Real Numbers

$$f(x) = \sum_{n=1}^{\infty} \frac{1}{1+n^2x}$$

- 1). Find $x \in \mathbb{R}$ for the series Converges absolutely
- 2) Find interval for the series Converges Uniformly
- 3) Find interval for the Converges Uniformly fails.
- 4). If the series converges, is f continuous? $\exists b \in \frac{1}{2}d \in \mathbb{R}$
- 5). " , f bounded?

Sol). Since $\left| \frac{1}{1+n^2x} \right| \leq \frac{1}{n^2c}$ if $x > 0$, for all $x > 0$, it Converges absolutely.

75. Let

$$f_n(x) = \begin{cases} 0 & x \notin \left[\frac{1}{n+1}, \frac{1}{n}\right] \\ \sin^2 \frac{\pi}{x} & x \in \left[\frac{1}{n+1}, \frac{1}{n}\right] \end{cases}$$

Show that $\{f_n\}$ converges to a continuous map, but not uniformly.

Observe that: $x \in \left[\frac{1}{n+1}, \frac{1}{n}\right] \Leftrightarrow \frac{\pi}{x} \in [n\pi, (n+1)\pi]$

First, for any $x_0 \in \mathbb{R}$, $f_n(x_0) \rightarrow 0$ as $n \rightarrow \infty$. (If $x_0 \leq 0$, then clearly $f_n(x_0) = 0$, and if $x_0 > 0$, then for some $N \in \mathbb{N}$, $n \geq N \Rightarrow x_0 > \frac{1}{n} > \frac{1}{n+1} > 0$. Thus $f_n(x_0) = 0, \forall n \geq N$.)

But, for any $n \in \mathbb{N}$, $\sup |f_n(x) - 0| = 1 \neq 0$, thus it is not uniformly.

Meanwhile, see that

$$\sum_{n=1}^{\infty} f_n = \begin{cases} 0 & x \notin (0, 1] \\ \sin^2 \frac{\pi}{x} & x \in (0, 1] \end{cases}$$

Clearly, it converges absolutely, but not uniformly, because for all $N \in \mathbb{N}$,

$$\sup \left| \sum_{n=1}^{\infty} f_n - \sum_{i=1}^N f_i \right| = 1.$$

16. Prove that the series

$$\sum_{n=1}^{\infty} (-1)^n \cdot \frac{x^2 + n}{n^2}$$

Converges uniformly in any bounded interval but not absolutely for any x .

Proof. First, since $\sum_{n=1}^{\infty} (-1)^n$ is bounded and $\frac{x^2 + n}{n^2} \rightarrow 0$ as $n \rightarrow \infty$ for any x , mathematically

the Dirichlet's Test gives the series Converges. But Not absolutely:

$$\sum_{n=1}^{\infty} \frac{x^2 + n}{n^2} = \sum_{n=1}^{\infty} \frac{x^2}{n^2} + \frac{1}{n} \quad \text{diverges.}$$

Nevertheless, it Converges uniformly, because: for fixed interval $[a, b]$ and for any $N \in \mathbb{N}$,

$$\left| \sum_{n=1}^{\infty} (-1)^n \cdot \frac{x^2 + n}{n^2} - \sum_{n=1}^N (-1)^n \cdot \frac{x^2 + n}{n^2} \right| = \left| \sum_{n=N+1}^{\infty} (-1)^n \cdot \frac{x^2 + n}{n^2} \right| \stackrel{(*)}{\leq} \left| \sum_{n=N+1}^{\infty} (-1)^n \cdot \frac{(\max_{[a,b]} x)^2 + n}{n^2} \right| \rightarrow 0$$

therefore, Converges uniformly. (The inequality $(*)$ is given by difference of two adjacent terms)

17. Define a map on $\mathbb{R}$ as

$$f_n(x) = \frac{x}{1 + nx^2}$$

1. $\{f_n\}$ Converges uniformly.

2. $f'(x) = \lim_{n \rightarrow \infty} f'_n(x)$ except $x=0$.

$$\text{Proof. } f'_n(x) = \frac{(1 + nx^2) - 2nx^2}{(1 + nx^2)^2} = \frac{1 - nx^2}{(1 + nx^2)^2} = 0 \quad \text{if } x = \pm \frac{1}{\sqrt{n}}$$

$$f_n\left(\frac{1}{\sqrt{n}}\right) = \frac{1}{2\sqrt{n}} \rightarrow 0 \quad \text{as } n \rightarrow \infty \quad \text{clearly, therefore}$$

$$\sup |f_n| \leq \frac{1}{2\sqrt{n}} \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

$$\text{And, for } x \neq 0, \quad f'_n(x) = \frac{1 - nx^2}{(1 + nx^2)^2} \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

$$\text{But } f'_n(0) = 1, \quad \text{while } f'(0) = 0.$$

78. Let $\{x_n\}$ be a sequence of distinct points of (a, b) , and if $\sum |C_n|$ converges, then

$$f(x) = \sum_{n=1}^{\infty} C_n \cdot I(x - x_n) \quad (\text{as } x \leq b)$$

Converges uniformly and f is continuous except $\{x_n\}$.

Proof. Converges uniformly is given by the Weierstrass Test directly.

79. Let $\{f_n\}$ be a sequence of continuous functions which converges uniformly to f on E .

Prove that the statement:

If $\{x_n\}$ in E converges to $x \in E$, then $\lim_{n \rightarrow \infty} f_n(x_n) = f(x)$.

Proof. Given $\epsilon > 0$, there exists $N_1 \in \mathbb{N}$ s.t. $n \geq N_1 \Rightarrow \sup_E |f_n - f| \leq \frac{\epsilon}{2}$.

And, since f_n is continuous, there is $N_2 \in \mathbb{N}$ s.t. $m \geq N_2 \Rightarrow |f_n(x_m) - f_n(x)| \leq \frac{\epsilon}{2}$.

Now, for $n \geq \max\{N_1, N_2\}$,

$$|f_n(x_n) - f(x)| \leq |f_n(x_n) - f_n(x)| + |f_n(x) - f(x)| \leq \epsilon.$$

Converse?

10. Denote $\{x\}$ as the fractional part of the real number x . Consider the function defined on $\mathbb{R}$

$$f(x) = \sum_{n=1}^{\infty} \frac{\{nx\}}{n^2}$$

Find all discontinuities of f , and show that they form a countable dense set.

Show that f is Riemann-integrable on every bounded interval.

Proof. Since $\left| \frac{\{nx\}}{n^2} \right| \leq \frac{1}{n^2}$, f converges uniformly, by the Weierstrass.

And, for each $n \in \mathbb{N}$, $x \mapsto \{nx\}$ has discontinuity at $x = \frac{k}{n}$ ($k \in \mathbb{Z}$).

7.12. Let $g, f_n: (0, \infty) \rightarrow \mathbb{R}$ be Riemann-integrable on $[t, T]$ whenever $0 < t < T < \infty$.

If $|f_n| < g$, and $f_n \rightarrow f$ uniformly on every compact subset of $(0, \infty)$, and $\int_0^\infty g(x) dx < \infty$,

then
$$\lim_{n \rightarrow \infty} \int_0^\infty f_n(x) dx = \int_0^\infty \lim_{n \rightarrow \infty} f_n(x) dx.$$

Proof.

17. 20. If f is continuous on $[0,1]$ and

$$\int_0^1 f(x) \cdot x^n dx = 0 \quad \text{for all } n=0,1,\dots$$

Then $f=0$.

sol). By the Weierstrass Thm, there exists a sequence of polynomials $\{p_n\}$ s.t. $p_n \rightarrow f$.

And, since f and p_n are continuous on the compact set $[0,1]$, they are bounded.

Therefore $f \cdot p_n \rightarrow f \cdot f = f^2$. Now, clearly $f \cdot p_n \in \mathcal{P}$, thus

$$\lim_{n \rightarrow \infty} \int_0^1 f(x) \cdot p_n(x) dx = \int_0^1 \lim_{n \rightarrow \infty} f(x) p_n(x) dx = \int_0^1 [f(x)]^2 dx$$

But by the assumption, for each $n \in \mathbb{N}$, $\int_0^1 f(x) p_n(x) dx = 0$.

Therefore, $\int_0^1 [f(x)]^2 dx = 0$, thus $f^2 = 0$, by $f^2 \geq 0$. Consequently $f=0$.